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Mapping Technological Knowledge Evolution in Chinese Universities

DOI: *leave as is, will be created by the journal*

Abstract

Introduction. Universities with equal patent output can differ in how they source, combine, and pass on technical knowledge. Patent counts do not show this difference; the relations recorded in patent documents do. This study proposes a framework for reading Chinese university patents as dated relational documents and for tracing how institutional knowledge structures change.

Method. Three relation layers—IPC classification, title–abstract text, and citations—are kept separate. Four process dimensions (acquisition, recombination, emergence, and reuse among observed citers) describe a university–period profile, each dated by the window in which it becomes observable. Transformation is assessed independently as sustained change in a university’s technological composition, tested against a within–university null.

Analysis. A fixed–layout technology map with period overlays, profile distributions adjusted for volume and field, and ordered trajectories combined with the transformation test constitute the analysis. The design states which measures a university–only patent and citation corpus can support and excludes reuse depth and economy–wide diffusion from primary claims.

Results. The paper provides the operational definitions and reporting design; the source database has not yet been analysed, so no profile value, trajectory count, or transformation event is reported as a finding.

Conclusion. The framework links knowledge organization and patent mapping to longitudinal institutional analysis while keeping each in–

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ference traceable to the patent relation that supports it and to the boundaries of the available data.

Introduction

Two universities with the same number of patents can produce them through different knowledge processes: one draws on a narrow set of external sources and refines an established core; the other connects technical elements that rarely appear together and is later reused by organizations outside its own field. Patent counts and citations record the output of both institutions but not the difference between them (Griliches, 1990; Hall et al., 2005; Nagaoka et al., 2010). Understanding how institutional knowledge structures change therefore requires reading patents as relational documents—linked to classifications, texts, earlier references, later citing records, and dates—rather than as units to be counted.

Information science already treats documents this way. Bibliographic coupling and co-citation identify intellectual connections (Kessler, 1963; Price, 1965; Small, 1973); knowledge maps represent domains and their change over time (Börner et al., 2003; Chen, 2006; van Eck & Waltman, 2010); patent overlays extend these methods to technology (Kay et al., 2014; Leydesdorff et al., 2014; Rafols et al., 2010; Yan & Luo, 2017). What is missing is a way to move from a map, which shows where an institution is located at one moment, to a trajectory, which shows how its knowledge relations change and whether that change exceeds sampling and measurement noise.

This paper proposes such a connection for Chinese university patents. The central idea is simple. Patent relations of different kinds—shared classifications, textual proximity, and citations—express different knowledge relationships and become observable at different times. Keeping them separate, and dating each one, makes it possible to describe an institution by four process dimensions that are observable when an invention is published or shortly after (acquisition of external knowledge, recombination of technical elements, early emergence, and documented reuse), and to assess structural transformation independently, as a sustained change in the institution's technological composition across periods. The two research questions are:

- RQ1.** How can acquisition, recombination, emergence, and reuse be measured from patent relations so that each retains its own meaning and its own observation time?
- RQ2.** Which university knowledge profiles and trajectories does a university-only patent and citation corpus support, once patent volume, technological composition, and coverage are taken into account?

The contribution is a traceable link between knowledge organization and longitudinal institutional analysis: every profile value and every trajectory statement can be followed back to dated patent records and the specific relation that supports it. The paper specifies the framework, the measures, and the reporting design,

and states explicitly which parts a university-only corpus can and cannot support. Empirical values remain to be computed; no finding is asserted below.

Related work

Patents are selective records of knowledge production. Not all inventions are patented, quality varies, and legal and organizational strategies shape the record (Griliches, 1990; Nagaoka et al., 2010). Citations require particular care: examiner-added references weaken their reading as inventor learning (Alcácer & Gittelman, 2006), and knowledge-flow estimates depend on technological controls and comparison groups (Jaffe et al., 1993; Thompson & Fox-Kean, 2005). Patents are therefore treated here as documented technical objects with dated relations, not as complete histories of invention.

Three relation types are used in patent mapping. Classification systems organize technical categories (Leydesdorff, 2008); text exposes similarities that shared codes miss (Kelly et al., 2021; Sarica et al., 2020; Whalen et al., 2020); citations record sourcing and later use. Overlay maps place organizations in a common knowledge space (Kay et al., 2014; Rafols et al., 2010), and relatedness research links positions in that space to entry and exit (Kogler et al., 2013; Rigby, 2015). Lay-out movement, however, can arise from a changed sample or algorithm rather than from changed knowledge. This study keeps visual coordinates fixed and measures change in the underlying distributions.

The process vocabulary comes from innovation research. Absorptive capacity concerns the recognition and use of external knowledge (W. M. Cohen & Levinthal, 1990); exploration and exploitation (March, 1991) and search depth and scope (Katila & Ahuja, 2002) describe learning orientations; combinatorial and atypical-combination studies motivate measuring unusual connections separately from their later impact (Fleming, 2001; Uzzi et al., 2013; Verhoeven et al., 2016; Wang et al., 2017; Youn et al., 2015). Emergence is more than novelty: its attributes include growth, coherence, impact, and uncertainty (Rotolo et al., 2015), of which scientometric methods operationalize a subset (Small et al., 2014). First citation and later accumulation carry different temporal information (Huang et al., 2019). Multilayer network research shows why heterogeneous relations should not be summed into one adjacency matrix (Boccaletti et al., 2014; Kivela et al., 2014). The present framework draws on these sources but makes a narrower claim: patent references index documented sourcing, not capability; rare code pairs index unusual combination, not usefulness; later citing records index bounded reuse, not learning.

Framework: relations, time, and transformation

Technological knowledge evolution is defined here as change in the sources, combinations, recognized directions, and documented reuse of an institution's technical knowledge, together with sustained change in how that knowledge is organized. Patents are the observable objects; universities are the contexts in which

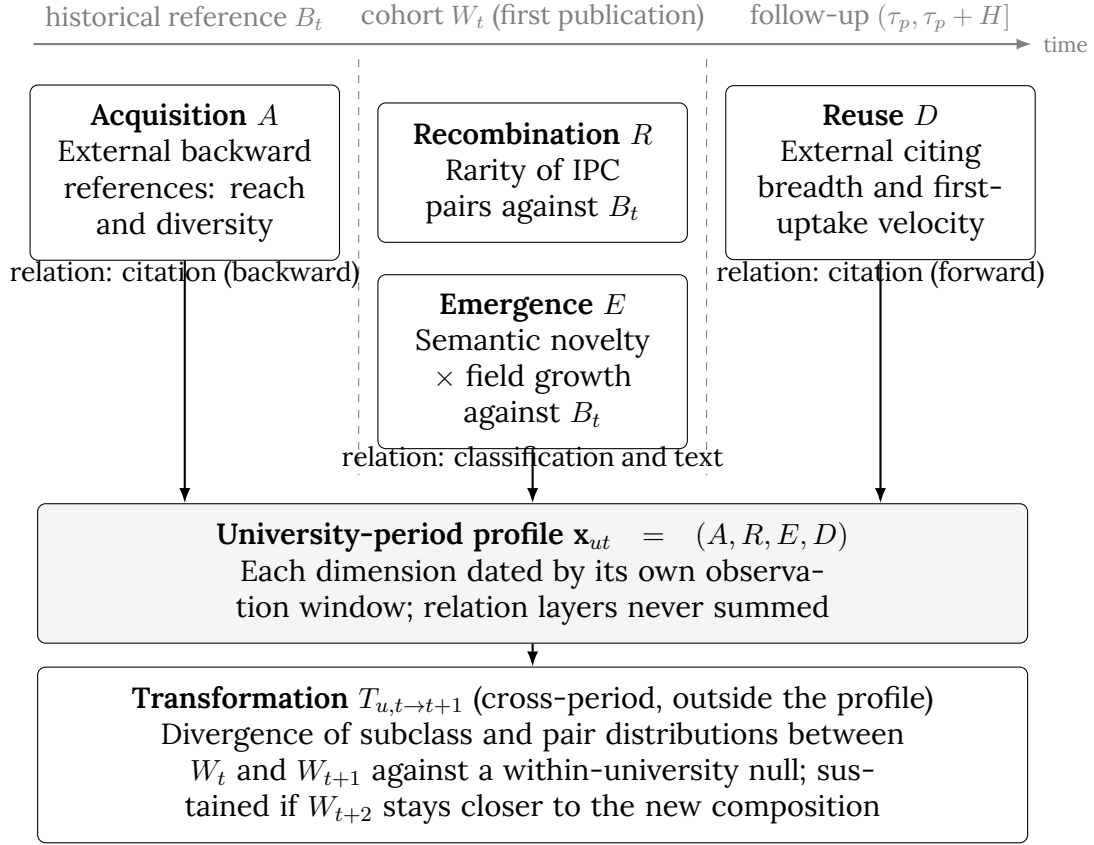


Figure 1. Framework. Four process dimensions are placed at the time they become observable and are drawn from distinct patent relations; transformation is a separate cross-period statistic, not a fifth input. No arrow denotes a causal effect or a compulsory stage.

they accumulate and connect. No patent or institution is assigned an intrinsic developmental type.

Figure 1 shows the design. Four dimensions describe a university-period profile,

$$\mathbf{x}_{ut} = (A_{ut}, R_{ut}, E_{ut}, D_{ut})^\top, \quad (1)$$

where A (acquisition) and R (recombination) are observable when inventions are published, E (emergence) uses the current cohort against its own historical reference, and D (reuse) requires a fixed follow-up horizon. Transformation, $T_{u,t \rightarrow t+1}$, is not a fifth input. It is a cross-period statistic on the distribution of technical categories and pairs, assessed separately with its own null. Keeping T outside the profile prevents a circular reading in which future structural change is used to describe a state and the state is then said to explain that change.

Two temporal rules follow. First, each dimension is dated by the earliest observation that determines it, and no indicator uses information published after its own observation window. Second, a profile at one period is a state, not a trajectory; trajectory statements require ordered profiles for the same institution and evidence that observed change exceeds a within-university null.

Four interpretive trajectory categories are retained as *hypotheses to be checked against these observables*, not as classes to be imposed. A *maintenance* trajectory shows a stable technical core (low T , no sustained coverage growth). An *expansion*

Coverage item	Value	Required definition or audit
Database/provider; extraction date	TBD	Frozen source and query manifest
Retrieval dates; analytic dates	TBD	First-publication cohorts; mature end date
Roster institutions; linked institutions	TBD	Dated roster and name crosswalk
Raw publication records	TBD	Before publication/application deduplication
Eligible invention units	TBD	Families, or applications if families unavailable
Eligible university-periods	TBD	Three-year non-overlapping cohorts
IPC subclasses; main groups	TBD	Version-harmonized vocabulary sizes
Usable title/abstract records	TBD	Count and coverage percentage
Backward-citation links	TBD	Resolved earlier public invention units
Forward-citation links within H	TBD	Unique family pairs, scope declared
External citing organizations	TBD	University-only or broader coverage
Excluded units by reason	TBD	Mutually exclusive first-failure counts

Table 1. Dataset scale and coverage, to be computed from the frozen source database. No value or date range in this table is an empirical estimate.

trajectory shows sustained growth in rarefied subclass coverage without unusual pairing. A *recombination* trajectory shows repeatedly high R supported by textual evidence of new element connections. A *transformation* trajectory requires T above its null in one transition and persistence of the new composition in the following period. An institution may follow different categories at different times, mixed cases remain visible, and the categories form no performance ranking.

Data and observable relations

Corpus and its boundaries

The intended source is the research team’s database of Chinese university patents with their citation records. Table 1 lists the coverage items that must be audited before analysis. Its rows determine which relations in the framework are observable and are therefore part of the design rather than a formality.

The primary cohort comprises published Chinese invention applications with at least one university applicant at first publication, regardless of later grant status; utility models and designs are excluded. Institutional coverage is defined by a dated university roster, with historical names, mergers, and affiliated hospitals resolved through time-specific linkage; ambiguous matches remain auditable rather than silently assigned. Publication and grant versions of an application are collapsed. If family identifiers are available, the invention unit is a simple family represented by its earliest eligible document; otherwise it is the deduplicated application, and the unit is declared. A patent with m_p university applicants contributes $\omega_{up} = 1/m_p$ to each; full counts are a sensitivity.

Three properties of a university-only corpus constrain the design and are stated here rather than discovered later. (i) Backward references from university patents are observable, but the cited documents’ own attributes are available only when the cited record is itself in the corpus or its classification is carried in the reference field. (ii) Forward citations to university patents are observable; the citing documents’ applicants are observable only where the citation record carries appli-

cant fields, and the citing documents' own forward links are generally not available. Reuse depth beyond one citation step is therefore excluded from the primary measure. (iii) Chinese invention patents receive few forward citations in their first year, so any indicator conditioned on early recognition will be zero for most records. Emergence is consequently measured in an early-information form that does not depend on recognition, with a recognized variant as a secondary measure.

Observation time

First publication is the main clock. $W_t = [b_t, b_{t+1})$ is a non-overlapping three-year cohort and $B_t = [b_t - 3, b_t)$ its historical reference; reuse uses a horizon of $H = 5$ years. Mature profiles require $b_{t+1} + H \leq d_{\text{extract}}$, and cohorts without a full historical reference are excluded. The number of mature cohorts this leaves is a property of the database, to be reported in Table 1; if fewer than three consecutive mature cohorts exist, trajectory analysis is restricted to description and the transformation test is reported as infeasible rather than approximated.

A university-period enters profile analysis with at least 20 attributed inventions and at least 80% usable attributed mass for the fields each dimension requires. Confirmed absence of citations is zero; an unavailable field is missing and never enters a denominator. For dimension j the eligible subset is P_{ut}^j with mass $M_{ut}^j = \sum_{p \in P_{ut}^j} \omega_{up}$, reported as a share of total mass.

Three relation layers

Figure 2 shows how the corpus is decomposed. The technology layer has IPC main-group nodes linked by fractional co-occurrence,

$$w_{ij}^{(t)} = \sum_{p \in W_t: m_p^C \geq 2} \frac{\mathbb{I}\{i, j \in \mathcal{C}_p\}}{\binom{m_p^C}{2}}, \quad i < j, \quad (2)$$

where $\mathcal{C}_p$ is the set of distinct main groups on patent p ; each multi-coded invention contributes one unit of pair mass. The semantic layer has patent nodes linked by cosine similarity of character 2–4-gram TF–IDF vectors of normalized titles and abstracts, with vocabulary and IDF fitted on B_t only (features in at least five and at most 80% of reference documents, capped at 100,000; log TF, smoothed IDF, L_2 normalization). The citation layer has patent nodes with directed edges from cited to citing document, duplicate family pairs collapsed, within-family links removed, and citing publication required to be strictly later. Applicant-overlap self-citations are excluded from the primary reuse graph. A patent–IPC incidence relation connects the layers; their adjacency matrices are never summed. Display graphs (normalized edges, top-10 neighbor union; mutual 15-nearest-neighbor semantic graph; Leiden communities, Traag et al., 2019) use fixed coordinates across periods; all indicators use unpruned weights.

Measures

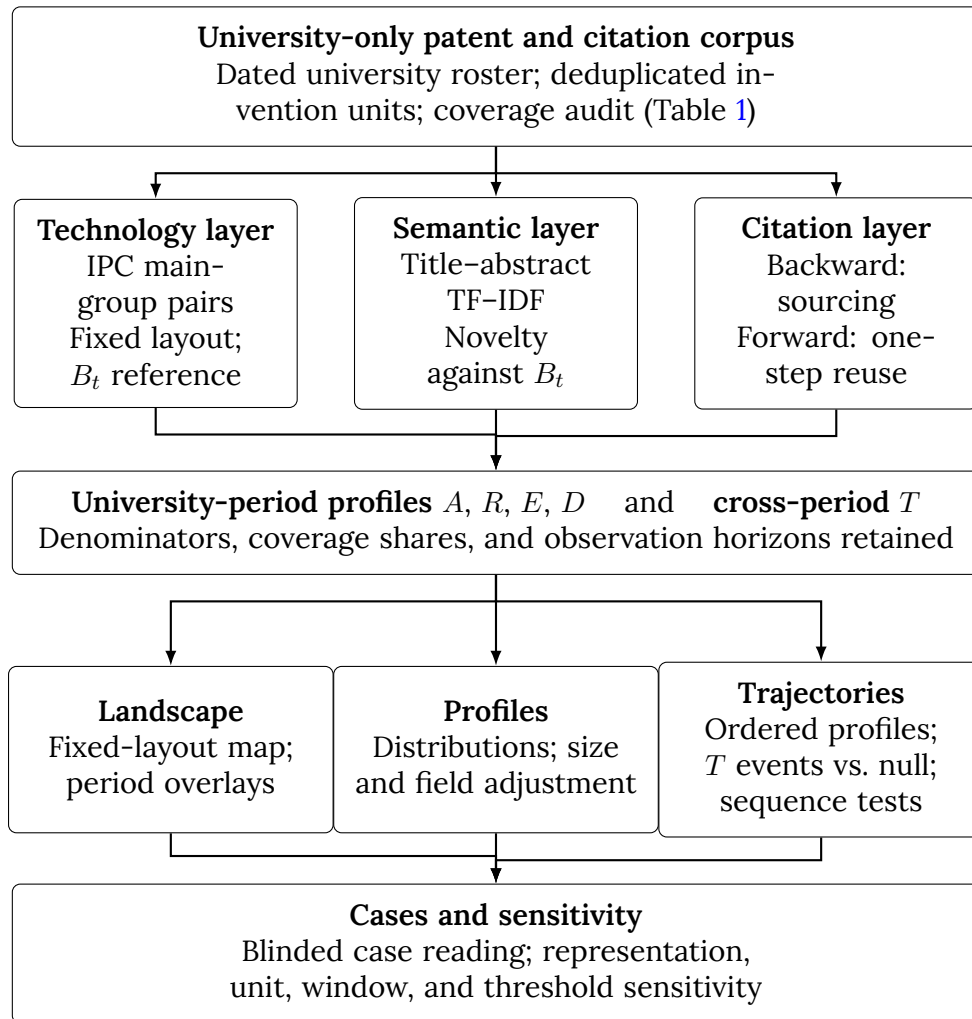


Figure 2. Data-to-inference workflow. Three relation layers are kept separate up to the profile; transformation is computed across periods from the technology layer alone.

Acquisition

Let q_{ut} be the attributed share of valid inventions with at least one resolved external backward reference, and let $\pi_{ut,k}$ be the fractional distribution of those references over a fixed, version-harmonized IPC subclass vocabulary of size K_t . Then

$$A_{ut} = q_{ut} \left[\frac{1}{2} + \frac{1}{2} \frac{-\sum_k \pi_{ut,k} \log \pi_{ut,k}}{\log K_t} \right], \quad (3)$$

so that confirmed absence of external sourcing gives $A = 0$ and the positive base-line distinguishes narrow sourcing from none. Both components are reported. A measures documented source reach and diversity (Shannon, 1948; Stirling, 2007), not absorptive capacity (W. M. Cohen & Levinthal, 1990).

Recombination

Let $c_{ij,t}^-$ be the historical fractional pair mass of main groups (i, j) in B_t , Q_t the number of possible pairs in the pre-established vocabulary, and $C_t = \sum_{i < j} c_{ij,t}^- > 0$. With $\alpha = 1/2$,

$$p_{ij,t}^- = \frac{c_{ij,t}^- + \alpha}{C_t + \alpha Q_t}, \quad r_{ij,t} = \frac{-\log p_{ij,t}^-}{-\log \{\alpha / (C_t + \alpha Q_t)\}}, \quad (4)$$

$$R_{ut} = \frac{1}{M_{ut}^R} \sum_{p \in P_{ut}^R} \omega_{up} \frac{\sum_{(i,j) \in \mathcal{L}_p} r_{ij,t}}{|\mathcal{L}_p|}, \quad (5)$$

where $\mathcal{L}_p$ is the set of valid mapped pairs on patent p . Unresolved codes map to one recorded out-of-vocabulary category; a patent with no valid pair is missing, not zero. Rarity indexes unusual combination, not novelty or usefulness (Fleming, 2001; Verhoeven et al., 2016; Youn et al., 2015). First-seen pair rates and degree-preserving permutations of the patent-IPC incidence provide complementary checks.

Emergence

Semantic novelty is $n_{p,t} = 1 - \max_{q \in B_t} s_{pq}$ over the full historical index, excluding same-family comparators. For subclass k , with fractional shares $\rho_{k,t}$ in the current cohort and $\rho_{k,t}^-$ in the reference,

$$g_{k,t} = \max \left\{ 0, \frac{\rho_{k,t} - \rho_{k,t}^-}{\rho_{k,t} + \rho_{k,t}^- + 10^{-12}} \right\}, \quad E_{ut} = \frac{1}{M_{ut}^E} \sum_{p \in P_{ut}^E} \omega_{up} n_{p,t} g_{p,t}, \quad (6)$$

where $g_{p,t}$ averages over the patent's subclasses. E is thus a mean of patent-level novelty-growth products: novelty and field growth must coincide in the same invention. It is an early-information measure available at publication. A recognized variant E^{rec} multiplies each product by an indicator of external citation within one year; given constraint (iii) in Section , it is reported with its nonzero share as a secondary measure and is not the primary emergence indicator. Neither variant covers the full emergence construct (Rotolo et al., 2015; Small et al., 2014); technical coherence is assessed through cases.

Reuse

Within $(\tau_p, \tau_p + H]$, let b_p count distinct external citing organizations where applicant fields are available and distinct external citing families otherwise (the level is recorded), and let $v_p = 1 - \Delta_p/H$ when the first external citation arrives after Δ_p years, with $v_p = 0$ for uncited patents. Each component is transformed to its midrank among positive values in the same primary subclass and cohort ($\phi(0) = 0$; back-off to IPC section and then cohort below 30 positives), and

$$D_{ut} = \frac{1}{2M_{ut}^D} \sum_{p \in P_{ut}^D} \omega_{up} [\phi^b(b_p) + \phi^v(v_p)]. \quad (7)$$

D measures *documented reuse among observed citers*. Because the citing side of a university-only corpus is incomplete, D does not measure economy-wide diffusion, and a two-step depth component is not part of the primary measure; it is computed as a sensitivity only where descendants' forward links exist. First uptake and accumulated breadth carry different information (Huang et al., 2019) and are also reported separately.

Transformation

Let $\mathbf{a}_{ut}$ and $\mathbf{r}_{ut}$ be the university's normalized subclass and pair distributions on shared vocabularies. Then

$$T_{u,t \rightarrow t+1} = \frac{\text{JSD}(\mathbf{a}_{ut}, \mathbf{a}_{u,t+1}) + \text{JSD}(\mathbf{r}_{ut}, \mathbf{r}_{u,t+1})}{2 \log 2} \in [0, 1], \quad (8)$$

with JSD the Jensen–Shannon divergence (Lin, 1991). A transformation event requires T above the 95th percentile of a within-university null that pools the two adjacent cohorts and permutes their period labels while preserving cohort sizes, confirmed under size-matched subsampling. A *sustained* event additionally requires the third period to be closer to the new composition than to the old and still distinguishable from the old. Mergers and IPC revision events are audited before any event is retained. T measures institutional reorganization of a portfolio, not a technological paradigm shift (Dosi, 1982; Funk & Owen-Smith, 2017); diversity is a distinct property (Porter & Rafols, 2009; Stirling, 2007).

Analysis and reporting design

The analysis proceeds in three steps, each answerable from the corpus described above.

Landscape. The technology layer is laid out once on the pooled corpus (Figure 3A); period overlays reuse the same coordinates (Figure 3B) so that entry, persistence, and exit of technical groups reflect the data rather than the layout. Coverage growth is evaluated against rarefied subclass coverage and portfolio size, since a larger sample alone makes more fields visible. Disagreement between the semantic and classification views marks cases for inspection.

Profiles. Distributions, zero shares, and denominators of A , R , E , D are reported for eligible university-periods (Table 2), together with pairwise correlations and their association with log patent volume, field composition, and period. Residual variation after these adjustments is the evidence that relational profiles add information beyond size and field. Profiles are then displayed on their original scales with whole-university bootstrap intervals (Efron, 1979), and universities are placed in profile space (Figure 4A). Model-based grouping of profiles (e.g., a Gaussian mixture with stability checks, Fraley and Raftery, 2002; Hennig, 2007) is a possible extension once the number of mature university-periods is known; it is not required for the questions posed here and is not asserted in advance.

Trajectories. For universities with consecutive eligible cohorts, ordered profiles are displayed alongside T and its null (Figure 4B). The four trajectory categories of Section are checked against their stated observable criteria; the counts of supported, mixed, and unsupported cases are reported. The proposition that sustained recombination precedes transformation is tested by comparing the observed frequency of “high- R followed by sustained T event” with its frequency under within-university order permutations that preserve each university’s profile values. A nonpositive or unstable contrast leaves the proposition unsupported.

Cases and sensitivity. Twelve trajectories sampled across profile regions and size strata are assessed by two technically qualified raters from dated patent evidence without model values; agreement, disagreements, and adjudication are reported (J. Cohen, 1960). Sensitivity covers invention unit, attribution, text representation (TF-IDF versus a frozen Chinese-capable encoder, Reimers and Gurevych, 2019), window length, citation horizon and self-citation scope, eligibility thresholds, and the E and D variants (Table 3). Common-sample agreement is reported separately from sample retention. All secondary associations are exploratory.

Results

Evidence status. Values are not yet computed; the paragraphs below fix what will be reported and contain no observed finding.

Coverage and landscape

The analytic population comprises [Pending: universities], [Pending: unique inventions], and [Pending: university-periods] in [Pending: mature cohorts]. Table 1 separates raw retrieval, deduplication, and eligibility, with the leading exclusion reasons [Pending: reasons and counts]. Of the citation layer, [Pending: share] of forward links carry citing applicant fields, which fixes the level at which b_p is computed. Figure 3 reports [Pending: node, edge, and community counts] and the retained mass; entry, persistence, and exit of technical groups are [Pending: observed patterns].

Profiles

Table 2 reports each dimension’s distribution and zero share. Correlations among A , R , E , D are [Pending: coefficients and intervals], and the share of variance remaining

A. Common technological space

Data required: IPC nodes and weighted pairs.

Position: one fixed reference layout.

Node area: fractional invention mass.

Edges: normalized co-occurrence.

Empirical map not yet computed.

B. Period overlays

Data required: period-specific IPC mass.

Reuse panel A coordinates.

Show entry, persistence, and exit.

Report retained mass and omitted nodes.

Empirical overlays not yet computed.

Figure 3. Technological knowledge space: analysis specification awaiting source data. The panels contain no synthetic nodes, invented clusters, or empirical coordinates.

Measure	Valid n	Mean	SD	Median	IQR	Zero %
Acquisition (A)	TBD	TBD	TBD	TBD	TBD	TBD
Recombination (R)	TBD	TBD	TBD	TBD	TBD	TBD
Emergence (E)	TBD	TBD	TBD	TBD	TBD	TBD
Recognized emergence (E^{rec} , secondary)	TBD	TBD	TBD	TBD	TBD	TBD
Reuse (D)	TBD	TBD	TBD	TBD	TBD	TBD
Transformation (T , per transition)	TBD	TBD	TBD	TBD	TBD	—

Table 2. Distribution of the process dimensions across eligible university-periods and of T across consecutive eligible transitions. Component summaries, missing counts, and coverage shares accompany each row in the output file.

after adjustment for volume, field, and period is [Pending: estimate]. Figure 4A shows where universities fall in profile space and which regions are populated; E^{rec} is nonzero for [Pending: share] of university-periods.

Trajectories and transformation

Among [Pending: count] universities with consecutive eligible cohorts, [Pending: count] show at least one T event above its null and [Pending: count] show a sustained event; [Pending: count] events are removed by the merger and IPC-revision audit. Trajectory categories supported by their observable criteria number [Pending: counts by category, mixed, unsupported]. The contrast between observed and permuted “recombination then transformation” sequences is [Pending: effect and interval]. Figure 4B displays sampled ordered profiles with their T values.

Cases and sensitivity

Rater agreement on [Pending: count] cases is [Pending: agreement and kappa]; principal disagreements concern [Pending: themes]. Table 3 reports the settings that change profile values or trajectory counts most; dependence on a single representation, on sparse citations, or on one eligibility threshold bounds the interpretation.

A. Profile space

Data required: A, R, E, D per eligible university-period.

Original scales; pairwise panels or a secondary PCA view for orientation.
Point size: attributed invention mass.
Whole-university bootstrap intervals.

No profile values computed.

B. Ordered trajectories

Data required: consecutive cohorts and T with null.

Horizontal axis: publication cohort.
Rows: prespecified sampled universities.
Bars: A, R, E, D ; markers: T above null, sustained events, audited mergers.

No trajectories assigned.

Figure 4. Profile and trajectory displays: analysis specification. Trajectory categories are read from the stated observable criteria and the independent T test, not from a fitted clustering.

Test family	Alternative and comparison	Observed result
Invention unit	Application versus family; common-sample profile change	TBD
Attribution	University-fractional versus all-applicant and full counts	TBD
Text representation	Character TF-IDF versus frozen Chinese-capable encoder; novelty rank agreement	TBD
Window length	Non-overlapping 2-, 3-, and 4-year cohorts; trajectory-category agreement	TBD
Reuse horizon and scope	$H = 3$ versus 5; self-citation exclusion; organization- versus family-level b_p	TBD
Eligibility	Minimum 10/20/30 inventions; completeness 70/80/90%	TBD
Emergence variant	E versus E^{rec} ; nonzero shares	TBD
Transformation null	Label permutation versus size-matched subsampling; event counts	TBD
Cases	Blinded rater agreement; cases rejecting the category reading	TBD

Table 3. Sensitivity and case results, pending computation. Each row reports the common-sample size, the change in profile values or trajectory counts, an interval where applicable, and the change in coverage.

Discussion

The framework's contribution to information science is representational. Each inference is tied to the relation that supports it: an unusual IPC pair supports a combination claim; semantic distance from a dated reference corpus supports a relative novelty claim; a later external citing record supports a bounded reuse claim; divergence between successive category distributions, against a within-university null, supports a reorganization claim. Retaining these origins makes agreement and disagreement among representations visible and connects patent mapping (Kay et al., 2014; Leydesdorff et al., 2014; Yan & Luo, 2017) to longitudinal institutional analysis without collapsing the relations into one score.

The design also states what a university-only corpus can support. It can describe sourcing, combination, early emergence, and reuse among observed citers, and it can test whether a portfolio's composition changed more than its own year-to-year variation. It cannot establish economy-wide diffusion, reuse chains beyond one step, or the causes of change. If recurring profiles or ordered sequences do not exceed their nulls, the framework remains a descriptive organization of evidence rather than an account of developmental mechanisms, and that outcome is reported as such.

Other limits persist. Patents omit scientific, tacit, and unpatented knowledge. Citations do not observe learning or contractual transfer (Alcácer & Gittelman, 2006). Novelty is relative to a corpus and a representation. Institutional events such as mergers, recruitment, and policy affect both profiles and later change. Profiles should therefore be read with their coverage and uncertainty, consistent with responsible use of research metrics (Hicks et al., 2015), and not as a ranking.

Conclusion

This paper connects the contents of patent records, their dated relations, and the time at which each relation becomes observable into a framework for describing how university knowledge structures change. Four process dimensions are measured from separate relation layers; transformation is assessed independently as sustained compositional change. The framework is specified against the actual boundaries of a university-only patent and citation corpus, so that its empirical claims—when computed—are limited to what the data observe. Extensions with enterprise citing records would allow reuse to be traced beyond the academic system, and publication–patent links would separate scientific from technological sourcing.

Data and code availability

The source database and computed outputs were not available for this revision; database permissions and any deposit must be verified by the authors. The accompanying LaTeX package contains the manuscript and display sources only. Prove-

nance, output fields, and audit requirements follow traceable data stewardship principles (Wilkinson et al., 2016).

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